Lecture 13

2/18/26

Fluorescence Microscopy (continued)

Start by reiterating strengths of fluorescence

1) Sensitive - can see a single fluorophore
2) Selective - can look at specific cellular components
   - comes at price of needing to label
     (antibodies a poor choice of labels, in Bethe's experience, due to nonspecific binding)
3) Contrast - from color w/ multiple labels
4) Living specimens - compatible w/ living samples
   - Cellular specimens
   - Patients (in medicine)
5) Can beat diffraction - new superresolution techniques can beat the diffraction limit by a factor of 10

Fluorescence also has negatives

1) Need to label
2) High light intensities
   - Photodamage - to specimen
   - Photobleaching - of fluorophore

These pros & cons reflect the mechanism of fluorescence

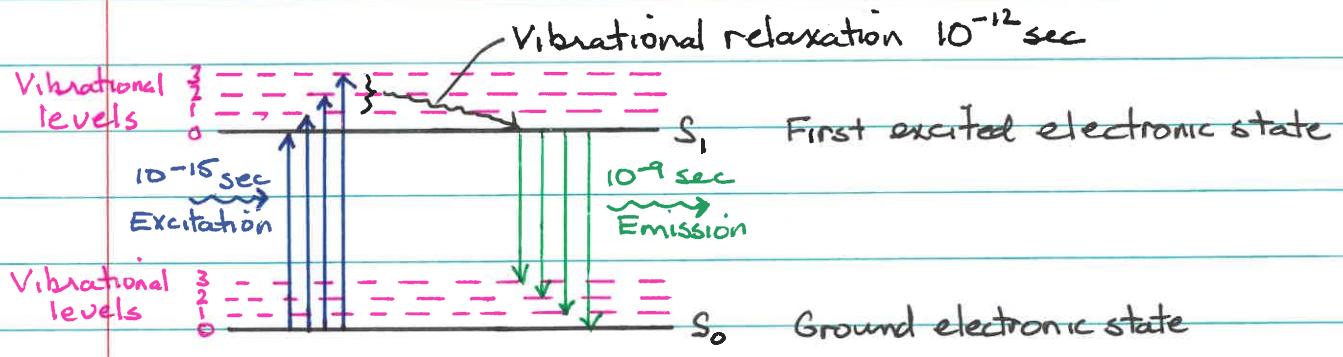
Mechanism of Fluorescence

The standard way to explain fluorescence is with a "Jablonski Diagram"

Molecules have quantized energy levels
Draw the levels in the diagram & plot the transitions that lead to absorption & emission

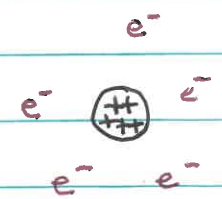
Different molecules have different energy levels
These differences lead to differences in the colors of light that different molecules absorb & that different molecules emit

The Jablonski Diagram shows the energy levels & transitions that play roles in Fluorescence



Why are they called electronic states?

They refer to the arrangement of electrons around the nucleus



Ground state - electrons closer to nucleus
Excited state - electrons farther from nucleus

Add energy, in the form of an excitation photon, to excite electrons into excited state

There also are vibrational levels associated w/ each electronic state. These correspond to different quantized vibrational states

Steps in Excitation & Emission

0) Fluorophore starts in lowest energy state

1) Excitation

Excite molecule w/ energy from light

Excitation, via absorption of excitation light, can take molecule into any vibrational level

Excitation takes $\sim 10^{-15}$ sec

2) Vibrational Relaxation

Excited molecule rapidly loses any excited vibrational energy as heat energy

Vibrational relaxation takes ~ 10^{-12} sec & occurs <u>before</u> light emission

3) <u>Emission (Fluorescence)</u>

Good fluorophores will lose excited-state energy via emission of light (i.e., fluorescence)

Emission always starts from lowest vibrational level of excited electronic state

Electron falls back into ground electronic state. It can land in any vibrational level.

Emission typically occurs about 10^{-9} sec after excitation ("the fluorescence lifetime")
Emission is a stochastic process

Excitation & emission can occur over a range of wavelengths because the energy of the photon required for excitation will depend on which vibrational level the electron is excited into & because the energy of the emission photon will depend on which vibrational level the electron falls into

Key Implications for Fluorescence Microscopy

- Excitation – fluorophore must be excited w/ light corresponding to the energy difference between S_1 and S_0

- Emission – emission light (fluorescence) will have lower energy than excitation light due to energy lost to vibrational relaxation

The energy difference between the higher-energy excitation light and lower-energy emission light makes it possible to distinguish* the emission light (i.e., the fluorescence) from the excitation light

* Fluorescence would be useless if not for this important fact

Quantification of Light Energy

The energy of light can be quantified

Einstein showed that light consists of tiny "particles" called "photons."

The energy of a photon is given by the following equation:

Energy of photon

$$\boxed{E_{photon} = h\nu}$$

Frequency of light

Planck's constant

h = Planck's constant
$= 6.63 \cdot 10^{-34}$ J·s

The frequency of light is related to its wavelength

Speed of light

$$\boxed{c = \lambda\nu}$$

Frequency of light

wavelength of light

c = speed of light = $3 \cdot 10^8$ m/sec

ν has units of cycles/sec (Hz)

The energy of a photon can be expressed in terms of wavelength using these two equations

$$\boxed{E_{photon} = h\nu = h\frac{c}{\lambda}}$$

The energy of a photon is inversely related to its wavelength

- Longer wavelength (larger λ) ⇒ lower photon energy
- Shorter wavelength (smaller λ) ⇒ higher photon energy

The fact that emitted photons have a different energy than excitation photons means they have different colors

DEMO Fluorescent rocks

Excitation: Ultraviolet — Higher energy
Emission: Green Orange — Lower energy

Green fluorescent protein (GFP) is another example of the color difference between excitation and emission. It is excited in the blue and emits in the green.

Stoke's Shift

The wavelength difference between the excitation light and the emission light is called the Stoke's shift.

The Stoke's shift reflects a change from higher-energy shorter-wavelength light to lower-energy longer-wavelength light.

The shift is from the blue end of the spectrum toward the red end of the spectrum.

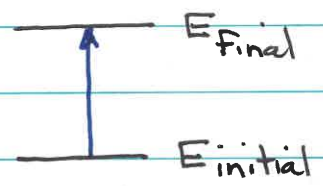
Bluer light → Redder light "red shift"

The Stoke's shift is SUPER important in fluorescence microscopy

Why is the Stoke's shift so important?

If you put a filter in front of the detector (e.g., the camera, the eyes, etc.) that blocks shorter-wavelength excitation light & transmits longer-wavelength emission light (fluorescence), you can get a fluorescence signal against a dark background

Want to be able to quantify the colors of absorbed & emitted light



$|E_{final} - E_{initial}|$ = energy that must come from photon

Excite - supply photon
Emit - get photon

Can calculate λ or ν of photon if know the quantized energy levels

$\boxed{\text{ANIMATION}}$ Mol Exp Jablonski Energy Diagram

Used animation to look at the following:

- Excitation, vibrational relaxation, emission
- Phosphorescence < Triplet state / Forbidden transition